

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Digital Forensics and Information Security –

Semester - II – April - 2026

Subject Code: CTMSDFIS24 SII P4

Date: 29/04/2026

Subject Name: Artificial Intelligence

Time: 2.30 PM TO 5.30 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

Q.1

Attempt any three.

Marks

(a) Explain the utility of Anaconda for environment management, and distinguish the primary roles of NumPy, Pandas, and Matplotlib in a data science workflow.

08

(b) A Security Operations Center (SOC) analyst is investigating a potential application-layer attack. They want to determine the relationship between the "Rate of Suspicious API Calls per Second" (X) and the machine learning Intrusion Detection System's "Threat Identification Latency in Milliseconds" (Y). Given the recorded sample data below, compute the Pearson Correlation Coefficient (r). Show your mathematical working and interpret the result.

08

X (Suspicious API Calls/sec): [40, 60, 50, 80, 70]

Y (Threat Identification Latency ms): [115, 145, 125, 175, 140]

$$r = \frac{N \sum xy - \sum x \sum y}{\sqrt{N \sum x^2 - (\sum x)^2} \sqrt{N \sum y^2 - (\sum y)^2}}$$

(c) Describe the typical pipeline for data handling and representation, emphasizing how missing values are treated and how data is visually summarized for insights.

08

(d) Define Machine Learning within the broader historical context of Artificial Intelligence (AI), and discuss the primary issues and challenges faced during the deployment of ML systems.

08

Q.2

Attempt any three.

(a) Contrast the structure of the biological brain with an Artificial Neural Network (ANN), and outline the mathematical components required to define and build a single-layer perceptron.

08

- (b) Differentiate between measures of central tendency (Mean, Median, Mode) and measures of dispersion (Variance, Standard Deviation) in summarizing a dataset. 08
- (c) Explain the necessity of partitioning a dataset into Training, Validation, and Testing sets, and discuss how a model's performance is evaluated using Accuracy and Loss metrics. 08
- (d) Describe the fundamental architecture of Convolutional Neural Networks (CNNs). Specifically, explain the operations of Convolutions and Pooling, and discuss how such models are fitted and evaluated. 08

Q.3

Attempt any three.

- (a) Define Long Short-Term Memory (LSTM) networks. Explain their specialized internal architecture and how building an LSTM model resolves the specific mathematical limitations of standard RNNs. 08
- (b) Analyze the phenomena of Underfitting and Overfitting in supervised machine learning, and explain the role of hyperparameters in mitigating these issues. 08
- (c) Compare and contrast Correlation and Regression, explaining their distinct roles in statistical analysis and predictive modelling. 08
- (d) Explain the mathematical purpose of activation functions in an Artificial Neural Network (ANN). Detail the characteristics, advantages, and limitations of any four commonly used activation functions. 08

Q.4

Attempt any two.

- (a) Compare and contrast Object Detection and Image Segmentation within Computer Vision. How do their outputs differ when analyzing a digital image? 07
- (b) Explain the significance of Context-Aware Text Analysis in NLP and how Text Visualization tools help human analysts interpret the results. 07
- (c) Describe how Machine Learning has revolutionized Malware Detection, specifically contrasting it with traditional signature-based detection methods. 07

Q.5

Attempt any two.

- (a) How does integrating Machine Learning into an Intrusion Detection System (IDS) improve its capability to secure modern IT environments? 07
- (b) Detail the process and importance of Text Vectorization in NLP. How does transforming text enable machine learning classifiers to analyze language? 07
- (c) Explain the concept of ML-driven Anomaly Detection in cybersecurity and how it operates to protect enterprise networks. 07

END OF PAPER

*TF-EDF
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